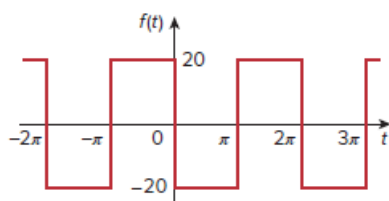


Problem

The function in Fig. 17.14 is half-wave symmetric.

- (a) True
(b) False

Figure 17.14:



Step-by-step solution

Step 1 of 1

Refer to Figure 17.14 in the textbook for a function, $f(t)$.

The Fourier series of the function is,

$$f(t) = -\frac{80}{\pi} \sum_{k=1}^{\infty} \frac{1}{n} \sin nt, \quad n = 2k - 1$$

Consider that the function to be half-wave symmetric, it should satisfy the following condition:

$$f\left(t - \frac{T}{2}\right) = -f(t)$$

Check this condition on the function, $f(t)$.

$$\begin{aligned} f\left(t - \frac{T}{2}\right) &= -\frac{80}{\pi} \sum_{k=1}^{\infty} \frac{1}{n} \sin n\left(t - \frac{T}{2}\right) \\ &= -\frac{80}{\pi} \sum_{k=1}^{\infty} \frac{1}{n} \sin n(t - \pi) \quad (\text{since, } T = 2\pi) \\ &= \frac{80}{\pi} \sum_{k=1}^{\infty} \frac{1}{n} \sin nt \quad (\text{since, } \sin(t - \pi) = -\sin t) \\ &= -f(t) \end{aligned}$$

Thus, the function $f(t)$ is half-wave symmetric.

Therefore, the correct answer is a.